

NGUYEN Trong Thang, BEng

🌐 Engineering Portfolio

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Employment History

- 2023 – 2026 **Research Assistant**, Power System Laboratory, Ho Chi Minh City University of Technology (HCMUT).
- 2022 – 2024 **Product & Brand Consultant**, Private Enterprise, Ho Chi Minh City.
Focus: Product research and development (R&D), customer consultation, brand advertising, and marketing strategies.
- 2020 – 2022 **Research Assistant**, Power Electric Research Laboratory, Ho Chi Minh City University of Technology (HCMUT).
- 2021 **Engineering Intern**, Medium Voltage Consulting Department, HCMC Power Engineering Construction Consulting Joint Stock Company.

Education

- 2019 – 2026 **BEng.Electrical Engineering, Ho Chi Minh University of Technology** Specialization: Power Systems
GPA: 7.07/10.0 for 133/132 Credits | Credits earned: 191/132
 - **Project 1:** Power Supply Design for a Brewery Factory - 9.0/10
 - **Project 2:** Unified Multi-View Clustering Model for Time-Series Data: Integrating Shape Invariance and Outlier Robustness - 9.5/10**Thesis title:** Proposing machine learning methods for classifying electricity customers with multiple characteristics - 9.7/10

Research Publications

- 1 T. N. Trong, K. P. Nguyen, and M. Q. Huynh, "Information-theoretic multi-view clustering for robust and interpretable analysis of electrical load profiles," in *2025 International Symposium on Electrical and Electronics Engineering (ISEE)*, 2025, pp. 232–237. [DOI: 10.1109/ISEE68370.2025.11223446](https://doi.org/10.1109/ISEE68370.2025.11223446).

Highlighted Software & Engineering Projects

- 2025 **Smart Grid Monitoring and Control System (Graduation Thesis)**
Technologies: Full-stack Web, AI/ML.
 - Developed a comprehensive web platform for power system dispatch, electricity market simulation, and AI-driven customer load profiling.
- ClusterLayoutOptimizer (Electrical Cabinet Layout)**
Technologies: K-Means Clustering, Geometric Algorithms.
 - Automated layout optimization system with obstacle avoidance and automatic wiring capabilities.

Highlighted Software & Engineering Projects (continued)

- **Physics-Informed Neural Networks (PINNs) for Power System Stability**
Technologies: Python, Deep Learning, Differential Equations.
 - Integrated physical laws into deep learning architectures to directly model and solve complex SMIB system dynamics.
- **Small-Signal Stability via Matrix Decomposition**
Technologies: Advanced Linear Algebra, MATLAB.
 - Formulated a Big-O optimized computational tool to block-diagonalize N-machine grid state matrices.
- **Optimal Control of Automatic Voltage Regulator (AVR) via LQR**
Technologies: LQR Algorithm, Control Theory.
 - Mathematically optimized the transient response and ensured robust voltage stability for synchronous generators.
- 2024 ■ **Golang-based SCADA System via Modbus Protocol**
Technologies: Golang, Modbus Protocol.
 - Engineered a real-time data acquisition and remote control system for industrial electrical equipment.

** Detailed source code, algorithms, and system architectures are available at my Engineering Portfolio.*

Skills & Expertise

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|-------------------------|--|
| Electrical Engineering | ■ Power System Control & Stability, SCADA & Modbus Protocol, High Voltage & Lightning Protection, Power Supply Design, IoT/Embedded. |
| Software & AI | ■ Machine Learning, Deep Learning (TensorFlow), Reinforcement Learning, Full-stack Web (React, Node.js), Linux, Git, Docker, SQL. |
| Programming | ■ Python, Golang, C/C++, Rust, Java, JavaScript, MATLAB, $\LaTeX$. |
| Engineering Tools | ■ ETAP, AutoCAD, Matlab Simulink, PSS/E, Revit (BIM MEP), Altium Designer. |
| Languages & Soft Skills | ■ English: Professional Vietnamese: Native French: Conversational
Academic Research, Technical Writing, Project Management, Team Leadership. |

Awards & Certification

- 2026 ■ **[Nominated] Best Graduation Thesis Award**, Faculty of Electrical and Electronics Engineering, HCMUT.
Recognized for achieving the highest evaluation score in the graduation defense committee.
- 2025 ■ **Semi-finalist, The 27th Euréka Scientific Research Student Award.**
Project: "A Big Data Analytics Tool Utilizing Machine Learning for Energy Consumption Insight and Optimization in the Power Industry".
- **Certified of Completion the Global Project-Based Learning Program.** Awarded by Prof. Goro FUJITA Shibaura Institute of Technology, Japan.

References

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